Brewing Under Pressure: More Realistic Tor Flow Correlation Considering Low FPRs

Haeseung Jeon*
haeseungjeon@ewha.ac.kr
Ewha Womans University
Seoul, South Korea

Yeonseo Cho*
sam.cho@ewha.ac.kr
Ewha Womans University
Seoul, South Korea

Nate Mathews
nate.mathews@mail.rit.edu
Rochester Institute of Technology
Rochester, United States

Rob Jansen
robert.g.jansen7.civ@us.navy.mil
U.S. Naval Research Laboratory
Washington, DC, United States

Marc Juarez
marc.juarez@ed.ac.uk
The University of Edinburgh
Edinburgh, United Kingdom

Se Eun Oh*†
seoh@ewha.ac.kr
Ewha Womans University
Seoul, South Korea

Abstract

Flow correlation attacks threaten Tor's anonymity by linking both ends of a communication circuit. Yet, existing correlation models rely on artificial datasets, resulting in significant performance degradation when evaluated against the sparse and highly variable nature of live Tor traffic. In this paper, we challenge this unrealistic evaluation by introducing two new datasets: Correlated GTT23, constructed from genuine exit traces, and Correlated CellFlow, comprising ground-truth flow pairs that we directly collected from the live Tor network. To tackle the unique challenges of these datasets, we propose ESPRESSO, a robust flow correlation attack tailored for realistic Tor traffic. ESPRESSO utilizes a density-based window representation via Kernel Density Estimation (KDE) to dynamically adapt to differences in transmission volume across a trace. Our results show that ESPRESSO outperforms state-of-the-art Tor flow correlation attacks, significantly improving detection accuracy across various scenarios, including standard settings and traffic drifted by temporal and network environment shifts. Notably, ESPRESSO achieves an impressive 88% True Positive Rate (TPR) at a False Positive Rate (FPR) of 10^{-5}, whereas DeepCoFFEA, the state-of-the-art attack, degrades to a near-zero TPR for the same FPR. This robustness holds even under adversarial constraints, including partial trace observations (as few as 500 cells) and limited visibility over the network under large numbers of concurrent connections (base rates as low as 10^{-4}). These findings indicate that flow correlation remains a severe, imminent threat, establishing both our datasets and ESPRESSO as the new standard benchmarks for evaluating future flow correlation attacks and defenses.

CCS Concepts

• **Security and privacy → Network security**; • **Computing methodologies → Machine learning**; **Artificial intelligence**.

Keywords

Tor, Metric Learning, Traffic Analysis, Flow Correlation, Privacy

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1 Introduction

With four to eight million daily users and nearly 10,000 *relays*, Tor is the most popular anonymity network today [2, 29]. Tor's anonymity properties stem from its implementation of onion routing [11, 55]: clients construct *circuits* selecting three relays at random, then pad each message to a fixed length and wrap it in three nested per-hop layers of encryption. As a result, no single relay in the circuit can observe both the source and destination of the communication, decoupling the endpoints from the perspective of local network observers—not only the relays themselves, but also Internet infrastructure providers such as Internet Service Providers (ISPs).

However, Tor remains vulnerable to traffic analysis attacks that exploit metadata such as timing, direction, and volume. One of the most prominent traffic analysis attacks against Tor is an end-to-end flow correlation attack. In this attack, a powerful adversary, such as a state-level actor, a major ISP, or an entity operating malicious Tor relays [25, 34], observes traffic at both the entry and exit points of the Tor network and matches entry–side flows with the corresponding exit–side flows. Flow correlation attacks are devastating to Tor because (1) the adversary can determine both the source and the destination of matched flows, completely deanonymizing Tor users, and (2) they affect *all* Tor connections independent of the chosen destination or traffic contents, unlike website fingerprinting (WF) attacks, which target *specific* destinations and traffic patterns [8, 13, 22, 30, 38, 39, 44, 51, 52].

Recognizing the severity of this threat, researchers have developed increasingly sophisticated flow correlation models. Early statistical approaches, such as RAPTOR [54], were limited by their reliance on handcrafted features and long observation windows. The introduction of deep learning marked a significant advance: DeepCorr [33] demonstrated that Convolutional Neural Networks (CNNs) could learn discriminative timing patterns directly from

*These authors are affiliated with the Division of Artificial Intelligence and Software.
†Corresponding author.

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raw traffic, and DeepCoFFEA [35] further reduced computational complexity from $O(N^2)$ to $O(N)$ through metric learning and a window-based amplification strategy.

Limitations of Previous Work. A fundamental limitation of previous approaches lies in the *data* used for evaluation: although prior state-of-the-art (SOTA) models like DeepCorr and DeepCoFFEA sought real-world applicability, they were trained and evaluated on highly sanitized, artificial datasets collected under lab conditions. Their data collection methodologies relied on automated tools that implicitly assumed artificial user models and used a local proxy server configured between the Tor exit relay and the destination server to record "exit-side" traffic. These artificial methods have been shown to produce synthetic data that is unrepresentative of real Tor user traffic across multiple dimensions: they inherently fail to accurately model exit location diversity and the extreme sparsity, severe network jitter, and dynamic delays caused by multiplexed communications and network congestion in the live Tor network [8, 26, 48]. Consequently, models optimized for these artificial datasets are fundamentally ill-equipped to handle the noisy, irregular conditions of real-world Tor traffic.

More Realistic Evaluation of Flow Correlation Attacks. We directly tackle the artificial flow correlation evaluation frameworks of prior works. First, we leverage the real-world GTT23 Tor exit traffic dataset [22] and the CellShift entry←exit trace transduction methodology [19] to construct the *Correlated GTT23* dataset, a new SOTA dataset for evaluating flow correlation attacks in realistic conditions. Second, to validate our use of CellShift in producing Correlated GTT23, we create a ground-truth *Correlated CellFlow* dataset by operating our own entry and exit relays and measuring paired traffic traces captured simultaneously at both ends of the live Tor network. Comprising approximately 503,000 entry–exit flow pairs, our new datasets represent the largest and most realistic benchmarks for flow correlation research to date. Using these datasets, we reevaluate prior SOTA models (DeepCorr and DeepCoFFEA) and observe that their area under the curve (AUC) significantly degrades by 0.2-0.4 compared to evaluations on the DeepCoFFEA dataset. This indicates that existing architectures are not well-suited to the inherent sparsity of real-world Tor traffic.

Introducing the ESPRESSO Framework. To address the deficiencies of existing models, we propose **ESPRESSO**,[1] a novel flow correlation attack model specifically tailored for real-world traffic. Unlike DeepCoFFEA, which extracts features based on fixed-time windows, ESPRESSO introduces a **Dens**ity-based **Win**dow (**DensWin**) utilizing Kernel Density Estimation (KDE) to dynamically determine effective time windows. Compared to DeepCoFFEA's fixed-time window approach, DensWin more effectively captures the extreme sparsity and high-variance characteristics inherent to real-world Tor traces. Notably, DensWin alone improves the AUC by 0.042 and significantly increases TPR by 0.659 even at low FPR thresholds (10^{-5}). ESPRESSO also integrates DensWin with triplet loss, incorporating token mixing and hard negative mining strategies to build robustness against varying conditions.

[1] **ESPRESSO**: **E**nhanced **S**equential **P**rocessing for **R**eliable **E**nd-to-end **S**ignal **S**ynchronization and **O**bservation. The source code and dataset of ESPRESSO and Correlated CellFlow are available at this artifact repository.

We conduct an extensive evaluation of ESPRESSO while comparing it to the SOTA attack. We find that ESPRESSO significantly outperforms DeepCoFFEA across various experimental scenarios, including concept drift and various base rates (achieving AUC improvements of 0.814 at maximum). Importantly, ESPRESSO achieves an impressive 88% TPR at a low FPR of 10^{-5}. Under this realistic constraint, the prior SOTA, DeepCoFFEA, suffers a significant drop to a near 0% TPR, highlighting its limitations in realistic settings. Furthermore, ESPRESSO shows robust performance even when a limited number of cells are available and at low base rates, indicating that the attack is feasible in real-world conditions.

Contributions. In summary, our key contributions are as follows:

(1) We construct Correlated GTT23 and Correlated CellFlow, the largest and most realistic datasets for flow correlation research to date, and demonstrate a SOTA evaluation methodology.
(2) We propose ESPRESSO, which integrates triplet loss, token mixing, and hard negative mining with DensWin, a novel windowing strategy for more effectively capturing the sparsity and high-variance characteristics inherent to real Tor traces.
(3) We carry out an extensive evaluation of ESPRESSO, showing that it outperforms SOTA attacks across several scenarios, including closed-world settings with guaranteed matches for all flows, concept drift, limited data conditions, and (for the first time) open-world scenarios with large volumes of unmatched traffic.

2 Background

2.1 Flow Correlation Attacks on Tor

Flow correlation attacks aim to deanonymize users of anonymous communication systems by linking traffic observed at different points in the networks to the same underlying communications [25, 32, 34]. While such attacks have been studied across a range of systems, including mix networks [36, 37], Tor is the primary focus of our research due to its scale and prominence.

Statistical Approaches. Sun et al. [54] proposed RAPTOR, which performs asymmetric traffic analysis using Spearman's rank correlation over per-unit-time byte counts, enabling deanonymization even when only one direction of traffic is observed. While RAPTOR demonstrated high accuracy under favorable conditions, statistical approaches are limited in their ability to handle the timing variability introduced by network jitter, multiplexed communications, and packet fragmentation in real Tor traffic, and required impractically long observation windows (e.g., > 5 minutes). More recently, Lopes et al. [28] proposed SUMo, which uses a sliding subset sum algorithm with an efficient XGBoost classifier for volumetric flow similarity, targeting Tor onion service sessions specifically.

Deep Learning Approaches. Nasr et al. [33] proposed DeepCorr, a CNN that formulates flow correlation as binary classification over (entry, exit) pairs. While DeepCorr significantly outperformed statistical methods by learning discriminative timing patterns directly from raw traffic, its pairwise formulation incurs $O(N^2)$ complexity when training on N flows, limiting scalability. Moreover, as a binary classifier, DeepCorr learns decision boundaries specific to the training distribution, making it brittle under concept drift. Oh et al. [35] addressed the scalability limitation with DeepCoFFEA,

which replaces binary classification with a CNN-based embedding model that learns on triplets: anchor (A), positive (P), and negative (N). DeepCoFFEA produces embeddings in a shared metric space using triplet loss with semi-hard negative mining to enforce a minimum margin between A-P and A-N distances, reducing training complexity to $2N$ forward passes. The resulting embeddings are compared through an *amplification* process that partitions each into fixed-time windows and performs window voting, significantly reducing false positives.

Passive vs. Active Correlation. All of the above are *passive* attacks: the adversary observes traffic without modification. A complementary line of work studies *active* flow correlation through watermarking, in which the adversary deliberately perturbs traffic to embed identifiable signals that can later be detected for correlation [17, 18, 45, 46]. In this paper, we focus exclusively on passive flow correlation, which poses a more challenging detection problem but requires no active manipulation of the network.

2.2 Toward Realistic Flow Correlation Datasets

To evaluate realistic flow correlation attacks, accurate datasets of correlated (entry, exit) flow pairs are needed. The attacks described above are limited in that they were developed using artificial datasets that have been shown to be unrepresentative of real Tor user flows across multiple dimensions.

Artificial Dataset Limitations. First, automated tools implicitly driven by artificial user models were used to programmatically produce synthetic entry-side flows. This practice fails to accurately model the full range of user behavior present in the live Tor network (including browser versions, number of simultaneous tabs, and user activity), and it has been shown to lead to significant overestimates of attack performance [8, 26]. Second, proxy servers were inserted between the Tor exit relay and the destination server to collect the correlated exit-side flows: Nasr et al. used a SOCKS proxy combined with a TCP packet capture [33], while Oh et al. used an Ethernet capture [35]. This practice fails to accurately model exit location diversity and the sparsity, jitter, and dynamic delays of the live Tor network. This single-proxy design was shown to introduce timing artifacts that inflate estimated attack performance: replacing a single US-based proxy with geographically distributed proxies yielded more realistic round-trip times and a statistically significant decrease in DeepCorr's correlation accuracy [48].

GTT23: Genuine Tor Exit Traffic. The first major step toward realism was taken in the measurement of the GTT23 dataset [22]: eight real Tor exit relays were run over 13 weeks to capture real Tor user traces reflecting natural user behavior. Compared to synthetic WF datasets such as BigEnough [31] and GoodEnough [40], GTT23 traces exhibit substantially shorter cell counts, higher intra-class variance, and a heavy-tailed distribution of inter-cell delays (ICDs). As Jeon et al. [23] noted, 29% of GTT23 traces exceed 100 seconds in duration, posing significant challenges for models optimized on denser synthetic data. These characteristics make GTT23 the most representative available source of real Tor traffic and a central building block for our evaluation.

Entry-Side Trace Generation. GTT23 contains only *exit*-side traces, but we also need correlated *entry*-side traces in order to evaluate flow correlation. Previous work proposed two methods for *transducing* each of a set of exit traces into its correlated entry counterpart. First, Retracer [21] leverages Shadow [20] to model realistic network conditions, replay exit traces during a high-fidelity network simulation, and record the corresponding (simulated) entry traces. Second, CellShift [19] estimates each circuit's round-trip time from Tor protocol metadata (relay control commands, directions, and timing information) and applies timestamp adjustments to simulate a change in the vantage point of trace observation from the exit to the entry. Although both methods produce correlated (entry, exit) datasets, we employ CellShift because it was shown to be superior: its entry traces are closer to ground truth than Retracer's, its traces lead to higher performance in both training and testing scenarios, and it is more efficient by five orders of magnitude [19].

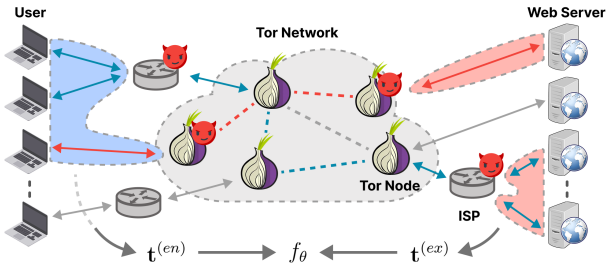


Figure 1: Flow correlation threat model.

Toward Realistic Flow Correlation Evaluation. Recent, concurrent work by Brighente et al. [7] has begun to address the gap between synthetic benchmarks and real-world conditions, evaluating flow correlation attacks against datasets collected using current Tor implementations and defenses, and finding that state-of-the-art attack effectiveness degrades in these settings. However, their evaluations share several of the methodological limitations identified above: they rely on lab-generated traffic from a small number of websites (50) rather than real Tor traces, they collect exit-side traffic through a single proxy server, and their models are trained using old datasets collected with different methodologies. These limitations leave open the central question of how flow correlation models perform on the distinct characteristics of real user traffic.

In this work, we directly address this question: we construct Correlated GTT23 using CellShift (Section 4.1) and collect a ground-truth Correlated CellFlow dataset from the live Tor network (Section 4.2), providing the most realistic benchmarks for flow correlation research to date.

3 Threat Model

Following prior work [33, 35], we consider a network-level adversary such as an ISP, an Autonomous System (AS), or a malicious Tor relay operator who conducts flow correlation attacks by observing traffic at both ends of a circuit (See Figure 1). Given a set of entry-side traces, denoted as $\mathbf{T}^{(en)}$, and exit-side traces, denoted as $\mathbf{T}^{(ex)}$, the objective is to identify *all* entry–exit trace pairs $\langle \mathbf{t}^{(en)}, \mathbf{t}^{(ex)} \rangle$ such that $\mathbf{t}^{(en)} \in \mathbf{T}^{(en)}$ and $\mathbf{t}^{(ex)} \in \mathbf{T}^{(ex)}$ correspond to the same circuit. We assume an attacker who leverages a trained attack model $f_\theta : \mathbf{t} \rightarrow \mathbf{e}$, which maps a raw trace $\mathbf{t}$ to a compact embedding representation $\mathbf{e}$. The similarity between an entry and exit traffic pair

is then evaluated using a distance metric $d(\cdot,\cdot)$ and a predefined threshold τ. Specifically, an entry trace embedding $\mathbf{e}^{(en)} \in \mathbf{E}^{(en)}$ and an exit trace embedding $\mathbf{e}^{(ex)} \in \mathbf{E}^{(ex)}$ are considered correlated if $d\left(\mathbf{e}^{(en)}, \mathbf{e}^{(ex)}\right) < \tau$ (details in Section 6.2).

We distinguish between two levels of adversarial visibility that lead to two classification settings: a *closed world* and an *open world*. In the closed world setting, the adversary has full visibility over the network and thus every trace in $\mathbf{T}^{(en)}$ has a corresponding match in $\mathbf{T}^{(ex)}$. In contrast, in the open world, the adversary can observe only a small fraction of the links, so $\mathbf{T}^{(en)}$ and $\mathbf{T}^{(ex)}$ may each contain traces with no counterpart in the other set. The closed world is an idealized setting and serves as a baseline for assessing attack performance. The open world is a more realistic setting and corresponds to local adversaries, such as ISPs and those that compromise a small fraction of the relays.

4 Datasets for Flow Correlation

One of our main contributions is *Correlated GTT23*, a Tor traffic dataset for evaluating flow correlation attacks that represents real Tor user behavior. Given its realism, this dataset enables a more accurate assessment of the performance the attacks would have on the live Tor network, allowing us to better understand the threat they pose to Tor users. Collecting data from real Tor users poses privacy and ethical concerns, especially for traffic on the entry side. To address these concerns, we leverage GTT23, a real Tor user traffic dataset collected at the exit side, and then apply *trace transduction* [19] to reconstruct the corresponding entry-side traffic as the adversary would have observed it. To validate the transduction method and ensure that it did not introduce any artifacts or bias into our evaluation, we collected *Correlated CellFlow*, an additional entry–exit live traffic dataset using our own Tor clients. In the remainder of this section, we describe in more detail the construction and collection of these datasets.

4.1 Correlated GTT23

We obtain a set of correlated (entry, exit) trace pairs from the exit traces in GTT23 using the open-source implementation of CellShift, and in particular, the TraceMove algorithm [19, § III. C.].

Conceptually, the algorithm first estimates a circuit round-trip time (cRTT) for each cell in the trace using the Tor protocol metadata that is available across the trace. Since a client⇆exit round-trip contains 6 hops, the per-hop latency is estimated as $\Delta = \frac{cRTT}{6}$. Then, cell timestamps are adjusted by the entry↔exit latency to simulate a shift in the vantage point of observation from exit to entry. The entry←exit cell timestamps are *incremented* by 2Δ (since the entry observes cells 2 hops *after* the exit sends them) and the entry→exit direction cell timestamps are *decremented* by 2Δ (since the entry sends cells 2 hops *before* the exit receives them). Finally, the cells are sorted by the updated timestamps (using a stable sort).

We apply CellShift to estimate the entry-side trace that would have been observed for each exit-side trace from GTT23. The result is a set of correlated (entry, exit) trace pairs that retain the real Tor user behavior patterns that are intrinsically embedded in GTT23. We describe how we use the traces for our evaluation in Section 6.1.

Table 1: Statistical comparison of the DeepCorr, DeepCoFFEA, Correlated GTT23, and Correlated CellFlow datasets. All values are means across all traces.

Metric	Dataset			
	DeepCorr	DeepCoFFEA	Corr. CellFlow	Corr. GTT23
Duration (*s*)	36	48	23	328
ICD Max. (*s*)	2	6	12	159
ICD Mean (*s*)	0.02	0.04	0.01	0.14
ICD Gini	0.73	0.91	0.93	0.99
Active Time Ratio (%)	0.73	0.44	0.31	0.23
Burst Count	349	235	135	120
Burst Size	5	13	33	32
Inter-Burst Gap (*s*)	0.06	0.14	0.10	0.96

4.2 Correlated CellFlow

We created *Correlated CellFlow*, a new dataset of correlated (entry, exit) traces that we will use to validate our primary evaluation results obtained with the more realistic Correlated GTT23. See Appendix B for more details about the measurement and results.

Measurement Process. To carry out the Correlated CellFlow measurement, we run an entry and exit relay in the live Tor network (in the US and EU) over a period of 10.5 weeks (2025-11-07 to 2026-01-21). The relays are configured with a 100 Mbit/s bandwidth rate and carry normal Tor traffic. We also run a Tor client (in the US), configured to pin our entry and exit relays as the first and last hop, respectively, in every circuit it builds. The client fetches each of the top 200k domains in the Tranco list[2] [27] through Tor Browser between 2025-12-20 and 2026-01-17 while our relays record traffic traces produced by our client. Correlated CellFlow is composed of the 169,432 successfully correlated (entry, exit) trace pairs.

Ethical Considerations. Prior to measurement, we submitted our plan and ethics analysis to the Tor Research Safety Board [1]. We received a "no objections" score and were given the go-ahead to proceed. More details are provided in Appendix C.

Realism. Our measurements were conducted on real Tor entry and exit relays, vantage points that an adversary could realistically obtain. This improves upon the exit-proxy methodology of previous work (see Section 2.2). However, because we are restricted to collecting our own traffic for ethical reasons, we still rely on artificial traffic generation tools. Compared with an ideal dataset of genuine user traffic, CellFlow therefore lacks natural diversity in user behavior (browsers and visited domains) and in circuit and relay selection, which may narrow the variance of the observed timing distribution. Thus, we limit our use of Correlated CellFlow to validating our primary results from Correlated GTT23.

4.3 Key Distinctions of Correlated Datasets

We analyze the key characteristics that distinguish real Tor traces from artificial traces to motivate the need for new flow correlation models that are designed explicitly for real-world deployment. We present a statistical analysis of traffic characteristics across DeepCorr, DeepCoFFEA, Correlated CellFlow, and Correlated GTT23 in Table 1. For this analysis, we uniformly subsampled 20,000 traces from each dataset.

[2]Generated on 2025-12-19, available at https://tranco-list.eu/list/ZWJ3G

Notably, Correlated GTT23 shows substantially longer flow durations, longer ICDs–the time interval between two consecutive cells–and larger inter-burst gaps. These longer durations stem from the data collection methodology: unlike other datasets, which stop capturing after a predefined duration, GTT23 terminates collection only upon circuit closure or once 5,000 cells have been captured. Furthermore, genuine user behaviors in Correlated GTT23, such as idle periods and asynchronous web interactions, lead to long ICDs and large inter-burst gaps. Finally, Correlated CellFlow exhibits burst counts and sizes closer to those of Correlated GTT23 than to those of the other datasets, suggesting that our newly collected dataset resembles realistic traffic.

To quantify the sparsity and ICD dispersion, we compute the Gini coefficient of the ICD distribution. The Gini coefficient equals 0 when all ICDs are identical and approaches 1 when the distribution is maximally skewed (i.e., extreme inequality). Network traffic inherently exhibits bursty behavior, consisting of dense bursts of activity interspersed with long idle periods, where ICD values are unevenly distributed. As shown in Table 1, while the mean ICD values are below 0.14 across all datasets, the maximum values are higher, yielding Gini coefficients above 0.73 in all cases. In Correlated GTT23, however, the coefficient becomes 1, indicating a heavy-tailed distribution where a few extremely high ICD values dominate the distribution.

To further investigate the proportion of these large ICDs, we define outliers as ICD values exceeding $\mu + 2\sigma$ (where μ is the mean and σ is the standard deviation). The fraction of the total duration occupied by these outliers is defined as the *idle time ratio*, while its complement is the *active time ratio*, reflecting the degree of bursty traffic. As shown in Table 1, the active time ratio drops from 0.73 in DeepCorr to 0.23 in Correlated GTT23. This confirms that the long durations in Correlated GTT23 are primarily driven by a small number of outliers. Moreover, unlike other WF datasets where large ICDs are distributed relatively evenly, Correlated GTT23 shows a heavy concentration of large ICDs either at the very beginning (within the first 10 cells) or toward the end of the trace, likely reflecting circuit establishment, user idle times, and the asynchronous loading of third-party content.

The heavy-tailed ICD distribution and the inherent sparsity driven by ICD outliers in real-world Correlated GTT23 traffic serve as the primary motivation for our proposed ESPRESSO architecture and DensWin feature representation, which we describe next.

5 The ESPRESSO Architecture

In this section, we present the ESPRESSO architecture and training methodology.

5.1 Overview

DeepCoFFEA's design has three limitations that become acute on realistic Tor traffic. First, it segments raw traces into fixed-time windows and feeds each window independently through a CNN, so no window can capture context beyond its own temporal span. Second, its fixed-time windowing produces many empty or near-empty windows in the presence of large ICDs, degrading feature quality and impeding training convergence. Third, the fixed-time representation itself is poorly suited to the heavy-tailed ICD distributions characteristic of genuine Tor traffic, as it cannot adapt to the extreme variation in transmission density across a trace.

ESPRESSO addresses these limitations through a new input representation and two tightly coupled architectural changes. We replace the per-window CNN with a CvT [58] backbone that processes the *entire* trace as a single global sequence, and we move the windowing operation *after* this backbone so that every window embedding is informed by the full temporal context of the trace. The third limitation is addressed by *DensWin*, a density-based feature representation that dynamically adapts bin boundaries to the transmission density of each trace.

Both ESPRESSO and DeepCoFFEA share the same high-level structure: separate Feature Embedding Networks (FENs) encode entry- and exit-side traces into a common metric space. ESPRESSO's FEN produces a sequence of window-level embeddings compared via cosine similarity at inference, preserving DeepCoFFEA's amplification framework. Figure 2 ① provides an overview. We provide additional details below.

Tokenization and Transformer Backbone. Our DensWin multi-channel tensor (described in Section 5.2) is projected into token embeddings by a 1D convolutional layer. The tokens pass through nine CvT-style transformer blocks, each combining (i) depthwise convolution for local feature extraction and spatial downsampling, (ii) Multi-Head Self-Attention (MHSA) as a *global* token mixer, and (iii) an MLP sublayer for position-wise mixing. Keys and values in each MHSA layer are produced via strided convolutional projection rather than linear projection, following CvT. Because every output token can attend to every input token, MHSA enables the model to relate temporally distant events, which is advantageous for GTT23's long-duration, bursty traces.

Post-FEN Windowing. Following DeepCoFFEA, we partition traces into windows for fine-grained cross-trace comparison and amplification. However, whereas DeepCoFFEA windows the raw input and processes each window independently, ESPRESSO windows the *learned feature stream*: a large-kernel convolutional layer aggregates contiguous segments of the transformer output, and a final linear layer projects these to the target embedding dimension. This post-FEN strategy confers three concrete benefits over the pre-FEN alternative: (1) each window embedding draws on a receptive field spanning the entire trace, capturing cross-window dependencies that pre-FEN windowing discards; (2) encoding the full trace in a single forward pass—rather than once per window—substantially reduces computational cost, making the larger transformer backbone practical; and (3) because DensWin produces a fixed number of mass-preserving bins regardless of trace duration, entry and exit window grids are aligned by construction, eliminating the temporal misalignment that would otherwise arise if DeepCoFFEA's fixed-time, per-packet representation were to be used.

5.2 DensWin Feature

This subsection outlines our feature extraction methodology and its underlying motivation. As identified in Section 5.1, fixed-time windowing degrades under the heavy-tailed ICD distributions of real Tor traffic. DensWin is specifically designed to address the extremely heavy-tailed ICD distribution and sparsity characteristic

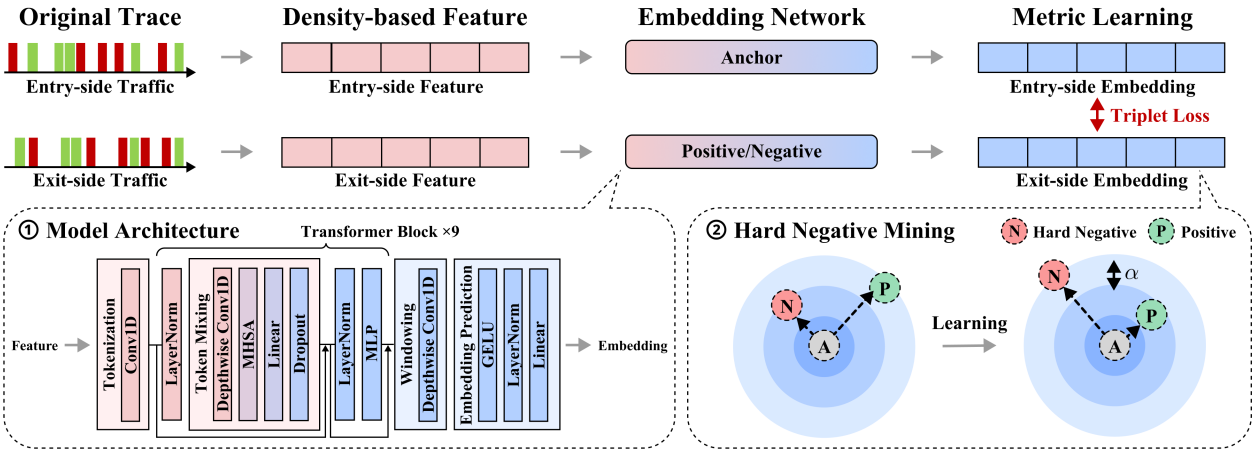


Figure 2: Overview of ESPRESSO.

of genuine Tor traces, which, as we show in Section 7.2, are the primary obstacle to maintaining high true positive rates at the low false positive rates required for operational feasibility.

Jeon et al. [23] were the first to address the unique characteristics of GTT23 traffic, such as high sparsity and ICDs, proposing a cell-count-based windowing technique for data augmentation. Oldenburg et al. [37] recently introduced density-based windowing for flow correlation attacks on the Nym mixnet, aiming to capture the "shape" of flows under differing transmission volumes between ingress and egress points. Building on these insights, we design a more effective feature representation for our attack.

The core intuition is that fixed-time bins allocate representational capacity uniformly across the trace duration, wasting most bins on idle periods while compressing dense bursts into too few bins and destroying the fine-grained structure that carries the correlation signal. DensWin instead partitions each trace into equal-probability-mass bins, concentrating bins where cells actually occur: narrow, high-resolution bins span dense bursts, while wide bins collapse idle gaps, ensuring that representational capacity is allocated in proportion to where discriminative information resides.

As illustrated in Figure 3, we use KDE to construct time bins that each carry approximately equal probability mass. Let $\{t_r\}_{r=1}^{n}$ denote the timestamp sequence of a single trace, sorted in increasing order. We first define m uniformly spaced grid points $\{g_q\}_{q=1}^{m}$ spanning $[t_{\min}, t_{\max}]$, with grid size $\Delta g = \frac{t_{\max}-t_{\min}}{m-1}$. Since the median total transmission time is approximately 100 seconds, we set m to 20,000, so that a grid interval corresponds to roughly 0.005 seconds for a trace.

$$\hat{f}(g_q) = \frac{1}{n}\sum_{r=1}^{n}\frac{1}{h\sqrt{2\pi}}\exp\left[-\frac{1}{2}\left(\frac{g_q - t_r}{h}\right)^2\right], \quad q = 1,\ldots,m \tag{1}$$

where $\hat{f}$ is the estimated probability density function (PDF) and h is the kernel bandwidth. Following Scott's rule, we set $h = \hat{\sigma} n^{-\frac{1}{d+4}}$, where $\hat{\sigma}$ is the sample standard deviation of the timestamps and $d = 1$ is the data dimensionality.

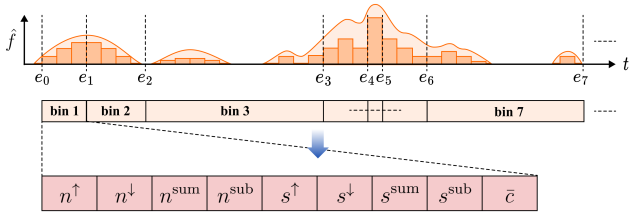


Figure 3: Overview of DensWin.

We then obtain a discrete cumulative distribution function (CDF) by cumulatively summing and renormalizing the KDE values:

$$C_q = \frac{\sum_{\ell=1}^{q}\hat{f}(g_\ell)}{\sum_{\ell=1}^{m}\hat{f}(g_\ell)}, \quad C_0 = 0. \tag{2}$$

Accordingly, C_q approximates the CDF of the KDE, truncated and renormalized over the observed time range.

To construct B bins, we define $B+1$ target cumulative probabilities as $u_b = \frac{b}{B}$ where $b = 0,\ldots,B$. For each u_b, we obtain the corresponding bin edge e_b in the time domain by linearly interpolating the normalized CDF values on the temporal grid. Consequently, each interval between two adjacent edges carries approximately $1/B$ of the probability mass estimated by the KDE.

Finally, we aggregate statistical features representing the number and size of the cells within each bin. Let $n^{\uparrow}$ and $n^{\downarrow}$ denote the number of cells in the client→exit and exit→client directions, respectively, with $n^{\text{sum}} = n^{\uparrow} + n^{\downarrow}$ and $n^{\text{sub}} = n^{\uparrow} - n^{\downarrow}$. We extract the corresponding directional size features $s^{\uparrow}$, $s^{\downarrow}$, s^{sum}, and s^{sub}. We additionally include $\bar{c}$, the average cumulative signed traffic within each bin.

This integration of timing, directional, and size representation enables robust feature extraction across both packet- or cell-level traffic datasets. Importantly, while traditional time-based binning significantly underperforms in the presence of long ICDs, our density-based approach dynamically adapts to irregular and sparse transmission volumes. An ablation study evaluating the impact of different bin and kernel types is provided in Section 7.5.

5.3 Metric Learning with Triplet Loss

Flow correlation is fundamentally a similarity task: the goal is not to assign flows to fixed classes but to map correlated flows from different vantage points onto nearby points in a shared embedding space. Like DeepCoFFEA [35], we train the FEN under a triplet metric learning objective, avoiding the $O(N^2)$ pair construction and distribution-specific decision boundaries inherent to the binary classification formulation used by DeepCorr [33]. Prior work in computer vision has additionally shown that such similarity-based objectives yield more robust representations under noisy and adversarial conditions than supervised classifiers [14, 43], which is key for the high-variance traces characteristic of real Tor traffic.

Triplet Objective. Following Hoffer et al. [16] and Schroff et al. [50], the triplet network learns embeddings such that correlated pairs (anchor, positive) are closer than uncorrelated pairs (anchor, negative) by at least a margin α (Figure 2 ②):

$$\|f(x^a) - f(x^p)\|_2^2 + \alpha \;<\; \|f(x^a) - f(x^n)\|_2^2, \tag{3}$$

where x^a, x^p, and x^n denote the anchor, positive, and negative inputs, respectively.

Online Hard Negative Mining. The negative sampling strategy is a critical design choice. DeepCoFFEA uses *offline* semi-hard mining, precomputing triplets over the full training set before each epoch. This approach has significant impacts on training costs, as mining complexity scales as $O((N/2)^2 \times W^2)$. We instead adopt *online batch-hard mining*, selecting the hardest negative within each batch on the fly. Online mining is both more adaptive, since batched mining tracks the evolving embedding space within an epoch, and is dramatically more efficient: it reduces training time per epoch from over 10,000 s to 33 s (Appendix D).

6 Experimental Settings

6.1 Datasets

To evaluate ESPRESSO under various experimental scenarios, we construct several subsets from the Correlated GTT23 and Correlated CellFlow datasets:

- **GTTCorr-All:** Randomly sampled exit-side traces from GTT23 weeks 1–13 with unique domains, partitioned into 10,000 trace instances for training and 9,000 for testing. Entry-side traffic is generated via CellShift. This is the primary dataset for standard evaluation (Section 7.2).
- **GTTCorr-DayN:** Same configuration as GTTCorr-All, but traces are sampled from the N-th day of GTT23, used to evaluate temporal drift (Section 7.3).
- **GTTCorr-NCells:** Same as GTTCorr-All, but restricted to the first N cells of each trace, used to evaluate the attack's feasibility under limited observation (Section 7.4).
- **CellFlow-All-GT:** Randomly selected ground-truth (entry, exit) pairs from CellFlow with unique domains, partitioned into 10,000 samples for training and 9,000 for testing. Used to validate CellShift (Section 7.1) and to evaluate network drift (Section 7.3).
- **CellFlow-All-CS:** Same configuration as CellFlow-All-GT, but entry-side traffic is replaced with CellShift-transduced traces. Used alongside CellFlow-All-GT for CellShift validation.

6.2 Evaluation Methodology

At inference time, ESPRESSO and DeepCoFFEA each produce a sequence of window-level embeddings for every trace. Given an entry trace $\mathbf{t}_i^{(en)}$ and an exit trace $\mathbf{t}_j^{(ex)}$, the models generate embeddings $\mathbf{e}_i^{(en)}$ and $\mathbf{e}_j^{(ex)}$, which are partitioned into W windows yielding segments $\mathbf{e}_{i,w}^{(en)}$ and $\mathbf{e}_{j,w}^{(ex)}$ for $w \in \{1, \ldots, W\}$. The core question is then how to aggregate per-window cosine similarities into a final correlation decision. We evaluate two complementary protocols:

(1) **Thresholding**: Following DeepCoFFEA [35], cosine similarity scores are computed for each window, and the top-k exit candidates are selected by ranking these scores. A window is regarded as correctly matched if its ground-truth exit trace appears within the top-k results, and a trace-level match is declared when more than 80% of its windows are correctly matched. For DeepCorr, which produces a single scalar prediction per (entry, exit) pair rather than window-level embeddings, we instead apply threshold sweeping over its prediction scores to compute the Receiver Operating Characteristic (ROC) curve.
(2) **MLP Classification**: Additionally, we propose a new MLP-based evaluation methodology that trains a four-layer fully connected MLP as a binary classifier over pairs of embedding windows; specifically, the pair $\langle \mathbf{e}_{i,w}^{(en)}, \mathbf{e}_{j,w}^{(ex)} \rangle$ is assigned a label of 1 if $i = j$, and 0 otherwise. The MLP-based evaluation yields a smoother ROC curve and often demonstrates improved correlation performance in our evaluations.

Under these evaluation settings, we report two metrics: (1) **AUC** for overall correlation detection rate and (2) **TPR@FPR**=10^{-n} to quantify real-world operational feasibility.

6.3 Training Settings

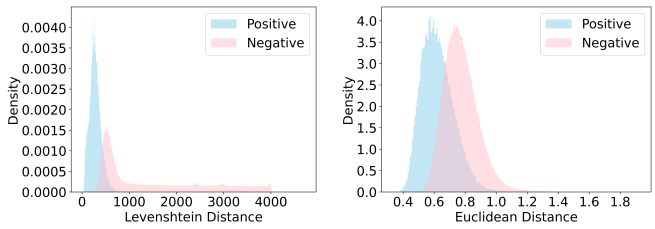
All experiments were executed on an Ubuntu 20.04 workstation with an NVIDIA RTX A6000 GPU (40 GB VRAM), 1 TB of RAM, 7 TB of SATA SSDs, and two NVMe SSDs. ESPRESSO was trained for 50 epochs using a batch size of 128, a learning rate of 10^{-3}, and a weight decay of 10^{-2}. We tuned the parameters such as the number of bins, kernel types, token mixer, and mining strategies, as detailed in Section 7.5. For DeepCorr and DeepCoFFEA, we use their reported hyperparameter configurations. For a fair comparison, we trained and evaluated all models from scratch.

7 Experimental Results

To comprehensively validate our proposed models and datasets, we structure our evaluation to address research questions as follows:

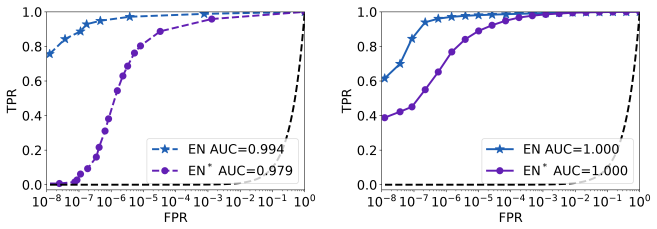
RQ1. Dataset Realism: *Does CellShift-transduced traffic reliably approximate ground-truth entry traces for flow correlation?* In Section 7.1, we compare ground-truth entry traces with CellShift-transduced entry traces and demonstrate that CellShift preserves the expected distance relationships between positive and negative samples, justifying Correlated GTT23's appropriateness as a flow correlation benchmark.

RQ2. Correlation Performance: *Can ESPRESSO outperform SOTA baselines and maintain its robustness against concept drift?* In Section 7.2 and Section 7.3, we benchmark the robustness of



(a) Levenshtein distance. (b) Euclidean distance.

Figure 4: Comparison of the distance between $\mathbf{t}_i^{(en)}$ and $\mathbf{t}_i^{(en^*)}$ and the distance between $\mathbf{t}_i^{(en)}$ and $\mathbf{t}_j^{(en^*)}$ in a histogram.



(a) Threshold results. (b) MLP results.

Figure 5: Comparison of the impact of CellShift-transduced and ground-truth entry-side training datasets (EN: CellShift-transduced entry-side; EN*: ground-truth entry-side. Black dashed line denotes random guessing).

ESPRESSO across standard settings, as well as temporal and environment shifts.

RQ3. Operational Feasibility: *How does strict real-world constraints impact the viability of ESPRESSO?* In Section 7.4, we explore practical constraints by evaluating the attack under time-limited observation windows and under various base rates.

RQ4. Architectural Impact: *To what extent do the individual components of ESPRESSO contribute to correlation detection capability?* We isolate and analyze the specific contributions of our density-based window features through an ablation study in Section 7.5.

7.1 Validating CellShift

Our primary evaluation dataset, Correlated GTT23, pairs genuine exit traces from GTT23 with entry traces generated via CellShift transduction (Section 4.1). Because the validity of all subsequent experiments depends on these transduced entry traces faithfully representing what an adversary would observe at the entry side, we must first establish that CellShift does not introduce artifacts that bias our evaluation.

To this end, we leverage the Correlated CellFlow (Section 4.2), which contains ground-truth (entry, exit) pairs captured simultaneously at both ends of the live Tor network. Correlated CellFlow allows us to directly compare CellShift-transduced entry traces ($\mathbf{t}_i^{(en)}$) against their ground-truth counterparts ($\mathbf{t}_i^{(en^*)}$). Specifically, we require:

§R1. $d(\mathbf{t}_i^{(ex)}, \mathbf{t}_i^{(en^*)}) \approx d(\mathbf{t}_i^{(ex)}, \mathbf{t}_i^{(en)}) > d(\mathbf{t}_i^{(en)}, \mathbf{t}_i^{(en^*)})$.
The CellShift-transduced trace should preserve the expected distance to the exit-side trace, while remaining closer to the genuine entry trace it approximates.

§R2. $d(\mathbf{t}_i^{(en)}, \mathbf{t}_i^{(en^*)}) < \frac{1}{N-1}\sum_{j\neq i}^{N} d(\mathbf{t}_i^{(en)}, \mathbf{t}_j^{(en^*)})$.
A CellShift-transduced entry trace must be closer to its corresponding true entry trace (positive pair) than to the average distance across all other entry traces (negative pairs).

We verify these requirements through both a direct distance metric analysis and a downstream attack performance comparison.

Distance Metric-based Evaluation. Table 2 reports the statistical distributions of Levenshtein, Manhattan, Euclidean, and Cosine distances across positive and negative pairs in the CellFlow-All-GT and CellFlow-All-CS datasets. For this analysis, we randomly sampled 50 negative pairs for each $\mathbf{t}^{(en)}$. Consistent with prior literature [19], we use directional features for these distance computations.

Table 2: Statistics between the ground-truth exit and entry sides and the CellShift-transduced entry-side traffic, computed based on directional features (P: positive, N: negative).

	Category	Levenshtein	Manhattan	Euclidean	Cosine
P	$d(\mathbf{t}_i^{(ex)}, \mathbf{t}_i^{(en^*)})$	360.23±141.76	0.25±0.08	0.70±0.11	0.25±0.08
	$d(\mathbf{t}_i^{(ex)}, \mathbf{t}_i^{(en)})$	299.10±126.30	0.22±0.08	0.65±0.11	0.22±0.08
	$d(\mathbf{t}_i^{(en^*)}, \mathbf{t}_i^{(en)})$	280.70±120.85	0.20±0.07	0.62±0.11	0.20±0.07
N	$d(\mathbf{t}_i^{(en)}, \mathbf{t}_j^{(ex)})$	1484.46±1126.12	0.30±0.10	0.76±0.12	0.30±0.10
	$d(\mathbf{t}_i^{(en)}, \mathbf{t}_j^{(en^*)})$	1489.40±1120.72	0.31±0.10	0.78±0.11	0.31±0.10
	$d(\mathbf{t}_i^{(en)}, \mathbf{t}_j^{(en)})$	1482.28±1127.84	0.30±0.10	0.77±0.12	0.30±0.10

Validating requirement **§R1**, the measured distance between the CellShift trace and its ground-truth entry trace constitutes the global minimum among all positive-pair distances. Furthermore, this distance remains strictly bounded below the distance of any negative pair, satisfying **§R2**. The distance distributions visualized in Figure 4 exhibit two distinctly separated modes, empirically demonstrating CellShift's effective reconstruction capabilities.

Attack Performance-based Evaluation. We extend our validation to attack performance by comparing ESPRESSO models trained on either CellShift-transduced (CellFlow-All-CS) or genuine (CellFlow-All-GT) traffic, with both evaluated on genuine traffic. Remarkably, as shown in Figure 5, training on CellShift-transduced traffic yields superior performance across all evaluation methodologies. This performance gain indicates that CellShift inherently acts as a robust data augmentation mechanism; the structural noise introduced during transduction effectively regularizes the model, compelling it to capture invariant, discriminative flow characteristics while avoiding overfitting to dataset-specific artifacts.

Summary. GTT23 captures genuine exit-side traffic from real Tor users but lacks entry traces. CellShift fills this gap: validation on Correlated CellFlow's ground-truth pairs confirms that the transduced entry traces preserve both the distance structure and the attack-relevant information of genuine entry traffic. We therefore use Correlated GTT23 as our primary evaluation dataset hereafter.

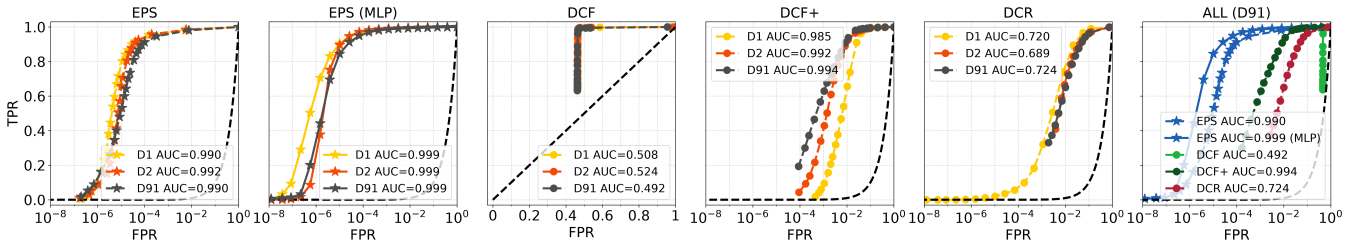
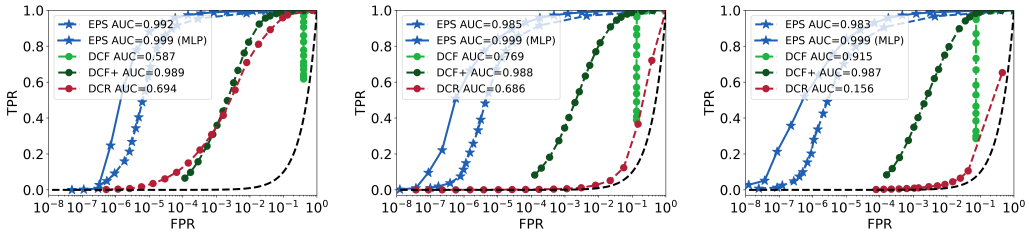


Figure 6: Thresholding performance of ESPRESSO and baselines under temporal drift. Trained with GTTCorr-Day1 and evaluated with GTTCorr-Day1 (D1), GTTCorr-Day2 (D2), and GTTCorr-Day91 (D91).



(a) Trained and evaluated with GTTCorr-All. (b) Trained with GTTCorr-All and evaluated with CellFlow-All-GT. (c) Trained with CellFlow-All-GT and evaluated with GTTCorr-All.

Figure 7: Thresholding performance of ESPRESSO and baselines, and MLP classification performance of ESPRESSO in (a): the standard setting with identical training and test datasets, and (b), (c): the network and long-term shift settings with different training and test datasets (EPS: ESPRESSO; DCF: DeepCoFFEA; DCF+: DeepCoFFEA+; DCR: DeepCorr).

7.2 Standard Correlation Benchmarks

Baseline Comparisons. We evaluated ESPRESSO against the baseline models using the GTTCorr-All dataset under a standard setting, without concept drift. As shown in Figure 7(a), ESPRESSO significantly outperformed both DeepCorr and DeepCoFFEA across both evaluation protocols. In the threshold-based evaluation, ESPRESSO achieves an AUC improvement ranging from 0.298 to 0.405, while the MLP-based evaluation yielded a near-perfect AUC of 0.999 (See Appendix E for the MLP-based evaluation of baselines). Notably, the ROC curves for DeepCoFFEA fail to cover the full threshold range (e.g., reaching TPR=FPR=0); its degraded estimations cause incorrect negative pairs to be persistently ranked within the top-k across multiple windows.

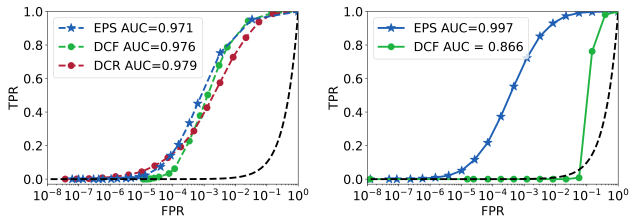
Importantly, ESPRESSO was the only model capable of maintaining a meaningful true positive rate at a low FPR of 10^{-5}. It achieved a TPR of 0.660 under the threshold-based protocol and 0.880 under the MLP-based evaluation, significantly outperforming DeepCorr (0.028) and DeepCoFFEA (0.000) in the standard threshold setting. This low-FPR capability underscores the practicality of ESPRESSO in real-world environments, where genuinely correlated pairs constitute a minuscule fraction of the overall traffic, making robust performance at near-zero FPRs a necessity.

Degraded DeepCoFFEA. A striking observation from our evaluation is that DeepCoFFEA suffers a severe performance drop on realistic data, falling significantly behind even DeepCorr—contradicting prior results on synthetic benchmarks [35]. We attribute this directly to DeepCoFFEA's fixed-time window strategy: for a traffic duration d, DeepCoFFEA statically defines windows of length t with an overlap d. When traffic is highly sparse like GTT23, this temporal rigidity causes severe feature degradation: numerous windows capture zero cells, while a sparse few become disproportionately saturated. This causes the model to converge poorly during training.

This issue can be addressed by using a more robust, density-based window strategy, similar to what was previously discussed in Section 5.2. We denote this improved design as DeepCoFFEA+. Specifically, we use the overlapping window strategy of the original DeepCoFFEA framework to isolate the performance impact of the segmentation criterion itself. Specifically, for a trace comprising n_t cells and a target window count of n_w, each window is dynamically sized to contain $\lceil n_t/n_w \rceil$ cells, with adjacent windows overlapping by a ratio r. We set $n_w = 11$ and $r = 0.5$.

As shown in Figure 7(a), DeepCoFFEA+ achieves a substantial AUC increase of 0.402 over the original DeepCoFFEA. However, its performance drops sharply when the FPR falls below 10^{-2}, whereas ESPRESSO maintains robust detection rates even at FPR $= 10^{-5}$. This comparison demonstrates two points: density- or cell-count-based representations yield a large gain over fixed-time windowing on traffic with large ICD gaps, and the architectural advances in ESPRESSO are necessary to sustain performance at operationally relevant FPRs.

Synthetic Dataset. To validate the robustness of ESPRESSO across datasets, we additionally trained and evaluated all models on the DeepCoFFEA dataset. To keep the configuration similar to that of Section 7.2, we used 10,000 pairs for training and 9,000 for testing; note that this split differs from the one used in prior work [35]. In this setting, we did not evaluate DeepCoFFEA+, since the use of



(a) Threshold results. (b) MLP results.

Figure 8: Performance of the ESPRESSO and baselines in the standard setting, trained and evaluated with DeepCoFFEA dataset.

proxy servers in the DeepCoFFEA dataset yields mismatched cell counts between entry- and exit-side flows.

Figure 8(a) compares ESPRESSO with the baselines. The performance of all models improves on this dataset, with DeepCoFFEA rising substantially from 0.587 to 0.976. Even so, ESPRESSO remains comparable to both DeepCoFFEA and DeepCorr. Moreover, as shown in Figure 8(b), ESPRESSO retains strong performance under MLP-based evaluation, whereas DeepCoFFEA achieves an AUC of only 0.866 and drops sharply for FPRs below 10^{-1}. This indicates that ESPRESSO is more robust than DeepCoFFEA across diverse evaluation settings.

Summary. The significant performance drop of existing state-of-the-art models on the GTT23 dataset reveals a major discrepancy between historical flow correlation benchmarks and real-world Tor traffic. Existing models struggle with the extreme sparsity and irregular delays of live Tor networks. In contrast, ESPRESSO maintains strong detection performance because its architecture and feature representation were explicitly designed for these conditions.

7.3 Evaluation against Drifted Flows

WF literature [26] establishes that attack efficacy degrades sharply after a 90-day temporal gap, due to evolving web content, dynamic third-party resources, and Tor infrastructure updates. Flow correlation research, however, has reported a different trend: DeepCoFFEA [35] claimed to sustain high correlation accuracy not only across temporal gaps exceeding one year but also under disparate network conditions, such as when the Tor circuit changes.

Given our earlier findings regarding the artificiality of prior benchmarks, a critical question remains: do these claimed resiliences hold under the unconstrained dynamics and extreme sparsity of realistic Tor traffic? To answer this, we evaluate flow correlation models against concept drift, incorporating both temporal and network condition shifts, to uncover their true operational robustness.

Short-Term Temporal Drift. To isolate the effect of temporal drift, we trained ESPRESSO on the GTTCorr-Day1 dataset and evaluated it on GTTCorr-Day2 and GTTCorr-Day91. The former simulates a practical scenario involving immediate attack deployment following data collection, whereas the latter represents a substantial delay between collection and deployment. Because GTT23 was collected over 13 weeks, the maximum achievable temporal gap within this dataset is approximately three months (91 days). By using only GTT23 data throughout, we ensure that performance is attributable to temporal shift rather than differences in network infrastructure.

Figure 6 shows the ROC curves. Across GTTCorr-Day1, GTTCorr-Day2, and GTTCorr-Day91, ESPRESSO (first) maintains highly stable performance with only moderate degradation in TPR at FPR $= 10^{-5}$ (0.772, 0.639, and 0.511). Conversely, the transition from Day 1 to Day 2 or Day 91 caused a slight decrease in AUC for both DeepCoFFEA (third) and DeepCorr (fifth); notably, DeepCorr exhibited a clear performance drop at FPR $< 10^{-3}$. DeepCoFFEA+ (fourth), like ESPRESSO, shows little to no change in AUC, indicating that metric learning approaches are intrinsically robust to temporal drift. Overall, ESPRESSO maintained strong performance even at very low FPR, whereas other models exhibit sharp performance degradation in the range of $0.5 <$ FPR $< 10^{-3}$ (last).

Robustness to Network and Long-Term Shifts. We next challenge our methodology with an extreme train–test mismatch by evaluating models trained on CellFlow-All-GT against GTTCorr-All, and vice versa. The former simulates a threat model where an adversary trains on data collected from a briefly operated, localized exit relay before deploying the model against the global Tor network. The latter represents an attacker leveraging a massive, public dataset (like GTT23) to launch an attack on their own specific infrastructure. This cross-evaluation imposes a compounded distributional shift encompassing both severe infrastructural discrepancies and a temporal gap of approximately three years (2023 for Correlated GTT23 versus 2025–2026 for Correlated CellFlow).

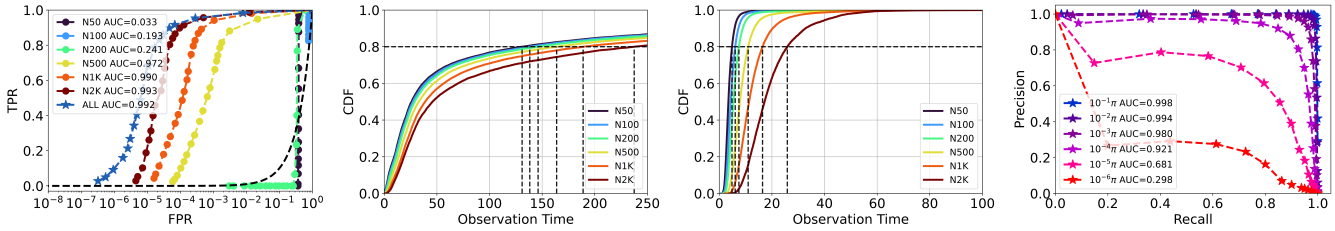
As shown in Figure 7(b) and 7(c), ESPRESSO demonstrates robustness against these severe distributional shifts in network conditions, outperforming both DeepCorr and DeepCoFFEA by a considerable margin of +0.068 to +0.827 in AUC. Furthermore, ESPRESSO exhibits highly stable performance under the MLP-based evaluation. DeepCorr, by contrast, suffers an AUC drop of up to 0.538 relative to the standard setting—a fundamental limitation of binary classifiers, which memorize decision boundaries tailored to the training distribution and fail under concept drift [35]. This highlights the necessity of taking a metric learning approach, as employed in DeepCoFFEA and ESPRESSO, for robust flow correlation.

Consistent with our standard setting evaluation, the enhanced variant DeepCoFFEA+ achieves better performance than its original version under this network drift setting. While DeepCoFFEA+ exhibits a rapid performance drop at FPR $< 10^{-2}$, ESPRESSO maintains superior detection capabilities even at a low FPR $= 10^{-5}$.

Summary. Our evaluations across temporal and infrastructural drifts highlight a clear divide: binary classifiers like DeepCorr degrade severely under distributional shifts, while metric learning approaches inherently resist temporal degradation. Building on this foundational robustness, ESPRESSO uniquely sustains high detection accuracy even under compounded infrastructural and temporal mismatches, confirming that it can maintain practical efficacy at low FPRs in long-term, real-world deployments.

7.4 Robustness under Real-World Constraints

In realistic threat models, adversaries operate under strict resource and observational constraints: they may observe only a fraction of the network, capture only a limited number of cells per trace, and encounter large volumes of unmatched traffic. Demonstrating that



(a) Threshold performance of the ESPRESSO varying number of cells during training. (b) CDF of the GTTCorr-NCells dataset. (c) CDF of the DeepCoFFEA dataset. (d) MLP performance of the ESPRESSO varying base rate during testing.

Figure 9: (a): Impact of the number of cells in the training dataset for ESPRESSO performance, (b), (c): CDF of the GTTCorr-NCells and DeepCoFFEA dataset, and (d): impact of the various base rates in the testing dataset for ESPRESSO performance.

ESPRESSO remains effective under these conditions is essential for establishing the practical feasibility of the attack.

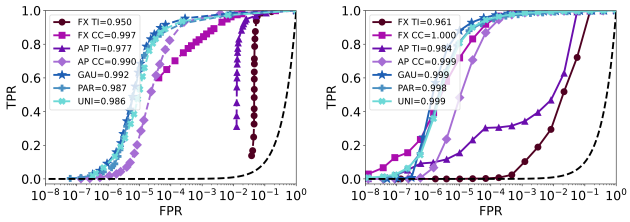
Varying Cell Counts. The number of cells an adversary can observe per trace directly determines the minimum monitoring effort required. We investigate the lower bound of cell counts needed to maintain adequate performance by training and evaluating ESPRESSO on the GTTCorr-NCells dataset, where $N \in \{50, 100, 200, 500, 1000, 2000\}$.

Figure 9(a) shows that capturing at least 500 cells per trace is required to achieve performance comparable to full traces. Below 100 cells, both AUC and ROC curves degrade substantially, approaching random guessing. Above 1,000 cells, performance reaches a near-perfect AUC, with only marginal differences from the full-trace baseline at FPR $< 10^{-4}$. These results demonstrate that just the first few hundred cells are needed for an effective correlation attack.

To translate cell counts into observation time, we computed the CDF of transmission duration for GTTCorr-NCells and estimated the time required to capture 80% of each trace's cell mass. Figure 9(b) presents these CDFs alongside the synthetic DeepCoFFEA dataset [35] in Figure 9(c) for comparison. Even for $N = 10$, GTTCorr-NCells requires over 100 seconds to reach the 80% threshold—significantly longer than DeepCoFFEA, which reaches the same threshold in under 30 seconds for $N = 2000$. For GTTCorr-500Cells, reaching 80% requires approximately 164 seconds, indicating that real-world observation windows are substantially longer than the 30–60 second windows commonly assumed in prior WF and flow correlation studies [31, 33, 35, 47, 52].

Varying Base Rates. As shown by Juarez et al. [26], evaluating traffic analysis under realistic base rates is imperative: models that appear effective under balanced conditions can degrade sharply in low-prior environments. To formalize this, we establish two levels of adversarial visibility that lead to two classification settings: a closed world and an open world. Prior flow correlation studies [33, 35] have operated under the closed-world assumption. In this setting, the adversary is assumed to have full visibility over the network, meaning every observed entry trace in $\mathbf{T}^{(en)}$ has a match in the exit traces $\mathbf{T}^{(ex)}$. Matching N entry and N exit flows yields N positive and $N(N-1)$ negative pairs, resulting in a base rate of $\pi = \frac{1}{N}$.

In practice, local adversaries, such as local ISPs or entities that compromise only a small fraction of the relays, operate in an open world. In this setting, the adversary can observe only a limited



(a) Threshold results. (b) MLP results.

Figure 10: Performance of the ESPRESSO with various bins and kernel types (FX TI: fixed time; FX CC: fixed cell count; AP TI: adaptive time; AP CC: adaptive cell count; GAU: gaussian kernel; PAR: parabolic kernel; UNI: uniform kernel).

fraction of the links, meaning $\mathbf{T}^{(en)}$ and $\mathbf{T}^{(ex)}$ inevitably contain significant volumes of unmatched traffic at both ends of the network. To simulate this realistic open-world constraint, we vary the proportion of matching (positive) pairs while holding the number of unmatched (negative) pairs fixed, and also fix positives while varying negatives, extending the base rate far below the artificial $\frac{1}{N}$ threshold. We benchmarked ESPRESSO across various base rates $\pi \in \{10^{-1}, 10^{-2}, \ldots, 10^{-6}\}$.

As shown in Figure 9(d), ESPRESSO maintains a high precision-recall AUC of 0.921 even at $\pi = 10^{-4}$, with noticeable degradation appearing only at $\pi \leq 10^{-5}$. This demonstrates that ESPRESSO can reliably perform flow correlation despite the presence of large volumes of unmatched traffic. These results also highlight a fundamental limitation of prior evaluation settings [33, 35, 54], which do not account for unmatched traffic. We advocate that future flow correlation evaluations should incorporate unmatched traffic conditions and highly imbalanced base rates as standard evaluation criteria to bridge the gap between laboratory performance and real-world applicability.

7.5 Ablation Study

Effect of Density-based Windows. To determine the most effective windowing strategy for realistic flow correlation, we compared two categories of approaches: non-density-based and density-based windows. For non-density-based windows, we evaluated four variants. *Fixed Time* segments each trace into windows of 0.03-second intervals, and *Fixed Cell Count* uses 10 cells per window. *Adaptive*

Time and *Adaptive Cell Count* divide the total transmission time or total cell count, respectively, into 2,000 equal windows per trace.

As shown in Figure 10, time-based windows (Fixed Time and Adaptive Time) yield the lowest AUC, as traces with high sparsity and large ICDs produce many empty windows. Cell-count-based windows perform better, with Fixed Cell Count even slightly outperforming density-based bins in overall AUC. However, this advantage is concentrated at high FPRs and diminishes at TPR@FPR=10^{-5}. Moreover, cell-count-based bins become unreliable when ingress and egress traffic volumes differ substantially, as observed in the DeepCoFFEA dataset. Density-based windows offer the most consistent performance across datasets. Among kernel choices, the Gaussian kernel outperforms the parabolic and uniform kernels at TPR@FPR=10^{-5} (0.660, 0.639, and 0.521, respectively), and we adopt it as our final configuration.

8 Discussion

In this section, we discuss the real-world implications of our results and other insights for future work.

Improving Tor's Privacy. Although our work is focused on Tor flow correlation *attacks*, our primary motivation is to inform the development of practical *defenses*. That is, we investigate Tor flow correlation attacks to better understand the threat they realistically pose to Tor and its users, with the hope that our work will inspire the future development and deployment of practical defenses that meaningfully improve privacy against realistic adversaries.

Circuit Compromise. A successful compromise of a Tor circuit depends on the adversary being in a position to observe both ends of the circuit. Using the bandwidth-weighted model of Johnson et al. [25], the probability of observing both the entry and the exit of a circuit is $P(O) \approx (B^{en}_{adv}/B^{en}_{total}) \cdot (B^{ex}_{adv}/B^{ex}_{total})$ for a *relay* adversary, or just $P(O) \approx (B^{ex}_{adv}/B^{ex}_{total})$ for a *network* adversary targeting a user it can already observe. Contributing more entry relay bandwidth B^{en}_{adv} or exit relay bandwidth B^{ex}_{adv} increases the probability of observation.

For a given (entry, exit) flow pair, let M denote the event that the pair is a true match, and F the event that the classifier flags it as correlated. Given the classifier's TPR and FPR and the base rate $\gamma = P(\mathrm{M})$, the threat can be characterized in two ways. First, the effective rate of compromise is $P(C) = P(O) \cdot P(\mathrm{F} \mid \mathrm{M})$, where $P(\mathrm{F} \mid \mathrm{M})$ is the probability that the classifier flags a correlated pair, i.e., its TPR. Second, whether an individual detection actually constitutes a compromise is governed by $P(\mathrm{M} \mid \mathrm{F})$, the positive predictive value: the probability that a flagged pair is correlated. By Bayes' rule, $P(\mathrm{M} \mid \mathrm{F}) = \frac{\gamma \cdot \mathrm{TPR}}{\gamma \cdot \mathrm{TPR} + (1-\gamma) \cdot \mathrm{FPR}}$, where the base rate γ is typically $1/N$, since only a single one of the N circuits active at any given time is the correct (entry, exit) match. Together, this formulation shows that greater bandwidth increases $P(O)$ and therefore the rate of compromise, whereas a higher base rate (i.e., fewer concurrent circuits) increases the confidence of each individual detection.

Realistic FPRs. The practical significance of a given FPR depends on the action the adversary takes upon a positive correlation. An adversary merely logging matches for later analysis might tolerate a high FPR, since false positives incur only storage costs; one dispatching law enforcement requires a far lower FPR to avoid costly, irreversible mistakes. The precise threshold at which a given adversary would act depends on their goals, resources, and tolerance for collateral damage, and remains an open question that future work should reason about explicitly [25]. China's Great Firewall (GFW) is instructive here: it escalates Shadowsocks probing incrementally over hours, with blocking decisions that (at the time) appeared to involve human oversight [3], and when blocking fully encrypted traffic it limits false positives by applying its heuristics to only about 26% of connections and restricting enforcement to IP ranges of popular VPS providers, yielding an estimated FPR upper bound of 0.006 [59]. Finally, an adversary monitoring a target over time can accumulate low-confidence correlations, with the joint FP probability decaying exponentially in the number of independent observations; even modest per-session TPRs therefore pose a meaningful long-term threat under sustained monitoring [56].

Potential Defenses against ESPRESSO. Numerous Tor path selection algorithms aim to reduce the risk that an adversary observes both sides of a flow: using trusted relays that are unlikely to be adversary-controlled [24], selecting entry and exit relays served by distinct AS networks [6, 53], choosing an entry close to the client [49], or selecting relays from a global view of the network [5]. None have been deployed, but our results help quantify the impact they may have against realistic adversaries and real-world traffic.

Padding is another commonly proposed defense. It injects dummy messages or timing delays so that client–middle traffic patterns differ from middle–destination patterns, complicating the ML task and raising FPRs. Although usually proposed against WF [31, 41, 42], padding has also been shown to degrade flow correlation [35]. To counter ESPRESSO's density-based windowing, future defenses could adapt Decaf [35], which injects peer-flow dummy traffic into randomly selected fixed-time windows. Scheduling such injections to create artificial high-density bursts could flatten the genuine transmission peaks and distort our KDE representation.

Traffic splitting distributes user flows across multiple entry relays [4, 10] or networks [15, 57], limiting the traffic observable on a given circuit but increasing the number of observable circuits. It may therefore increase the threat of flow correlation, particularly for models like ESPRESSO that are robust to low-volume flows. Conflux was deployed in Tor for performance reasons [12], but its effect on flow correlation attacks remains an open question.

9 Conclusion

We propose ESPRESSO, a metric learning framework with triplet loss that consistently outperforms SOTA flow correlation baselines across standard, temporal drift, and network shift settings. We also introduce Correlated GTT23 and Correlated CellFlow, the largest and most realistic benchmarks to date. On these, ESPRESSO achieves a strong TPR at very low FPR and remains effective under limited cells or correlated pairs, demonstrating the real-world viability of flow correlation attacks and the need for practical defenses.

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Appendices

A Open Science

The source code and dataset of ESPRESSO and Correlated CellFlow are available at this artifact repository.

B Extended Correlated CellFlow Description

Although Correlated GTT23 embeds real user behavior patterns, it is possible that the CellShift transduction introduces a data artifact that could bias our ML model. To identify such potential bias and validate any results we achieve with GTT23, we need a secondary dataset that is produced independent from CellShift. Thus, we created *Correlated CellFlow*, a new dataset of correlated (entry, exit) traces measured using our own Tor relays and client.

Measurement Process. To carry out the Correlated CellFlow measurement, we run an entry and exit relay in the live Tor network over a period of 10.5 weeks (2025-11-07 to 2026-01-21). The relays are run on different continents (US and EU), are configured with a 100 Mbit/s bandwidth rate and a 200 Mbit/s bandwidth burst, and are given time to become integrated into the network with high background traffic loads prior to measurement. We also run a Tor client in the US, configured to pin our entry and exit relays as the first and last hop, respectively, in every circuit it builds. Our nodes run a modified version of Tor v0.4.8.21 that integrates the measurement methodology of Cherubin et al. [9] in which the client uses a unique Tor cell type to signal to our relays (1) that the circuit is ours and is safe to measure, and (2) a string label for the trace. Our relays record labeled traces of the circuits created by our client.

With the measurement infrastructure in place, we create a Python script to automate web requests over Tor using the open-source stem and tbselenium packages. For each of the top 200k domains in the Tranco list[3] [27], the script starts an ephemeral Tor client process configured with a unique session ID, requests the URL `https://domain` through Tor Browser using its default *Standard* security level, waits 30 seconds, and then stops the client. The script runs from 2025-12-20 to 2026-01-17, recording errors while measuring 5 domains in parallel.

Measurement Results. In total, we requested 200,045 URLs with our Python script (with some URLs duplicated due to stopping and resuming). Tor browser reported 145,013 (72.5%) successes and 55,032 (27.5%) failures. The majority of the errors are network-related, with `dnsNotFound`, `netTimeout`, `connectionFailure`, and `nssFailure2` jointly accounting for over 95% of all cases. The application-level read timeouts comprise a relatively smaller fraction (3.5%), while other error types are minor (1.3%).

The relay-observed traces were collected from the entry and exit relays and, after discarding traces with fewer than 25 cells (likely failed requests), they were correlated according to the client-assigned labels. Correlated CellFlow is composed of the remaining 169,432 correlated (entry, exit) trace pairs. Note that attempting to fetch a URL may result in multiple circuits, e.g., if the browser is redirected or retries a request that failed mid-stream.

Because we use programmatic webpage fetching tools, the user patterns in Correlated CellFlow are synthetic and less realistic than those in GTT23. However, because each URL's trace is measured from *both* the entry *and* the exit, Correlated CellFlow contains (entry, exit) trace pairs as they were naturally observed without CellShift or other simulation methods. Thus, it serves as a useful dataset for validating our GTT23-based evaluation (See Section 7.1).

C Ethical Considerations

The Correlated CellFlow measurement described in Section 4.2 and Appendix B was conducted on the live Tor network. The Correlated CellFlow traces were of circuits created by our own client and measured by our own entry and exit relays. We did not record data of any circuit other than our own. The data we recorded was encrypted using a public-key encryption scheme in which the private key was stored offline, providing data security and preventing unauthorized disclosure in case the machines were compromised.

Our measurement involved Tor middle relays and Internet servers not under our control. We limited our measurement rate to 5 parallel URL fetches: these were balanced across middle relays following Tor's load balancing algorithm, and we fetched each URL only once. We believe that the bandwidth and CPU resources that our entry and exit relays contributed to the Tor network while they were online exceed the resources that we consumed from the external nodes during our measurement.

[3] Generated on 2025-12-19, available at https://tranco-list.eu/list/ZWJ3G

Table 3: Complexity comparison of ESPRESSO and baselines.

Metric	Model		
	DeepCorr	DeepCoFFEA	ESPRESSO
# of Param.	183.4M	2×2.28M	2×2.13M
DNN Runs	$O(N^2)$	$O(2N)$	$O(2N)$
Similarity Comp.	-	$O(N^2 \times W^2 \times D)$	$O(N \times W \times B \times D)$
Mining Comp.	-	$O((N/2)^2 \times W^2)$	$O(N \times B)$
Time/Epoch (s)	4,104	10,432	33

According to the original paper [19], the collection of GTT23 was designed to prioritize safety and privacy and was developed through consultation with the Tor Research Safety Board [1], which resulted in a "No Objections" score. Prior to measurement, we submitted our plan and ethics analysis to the TRSB. We also received a "No Objections" score and were given the go-ahead to proceed.

D Complexity Analysis on Baselines

In this section, we compare ESPRESSO with baselines across multiple metrics to demonstrate its efficiency.

Parameter Efficiency. As reported in Table 3, ESPRESSO remains the most lightweight model despite using two separate networks for A and N/P, requiring only 2% and 93% of the parameters used by DeepCorr and DeepCoFFEA, respectively. In contrast, DeepCorr jointly processes (entry, exit) traffic pairs with a binary classifier, leading to heavy reliance on fully connected (FC) layers and making it the most parameter-intensive model.

Complexity of DNN Runs. DeepCorr requires N^2 DNN runs for N flows, as it trains on both positive and negative (entry, exit) pairs. While DeepCorr sets the number of negative samples as a hyperparameter and limits it to 199, such an approach remains costly and introduces additional tuning overhead. In contrast, DeepCoFFEA and ESPRESSO employ a triplet network that embeds entry and exit flows separately, requiring only $2N$ DNN runs.

Complexity of Data Mining. DeepCoFFEA performs semi-hard mining by examining all data at the beginning of each epoch. Here, the (entry, exit) flow embedding of DeepCoFFEA contains a total of NW windows, where each of the N samples has W windows of D dimensions ($W = 11$ and $D = 64$). Since each window is split and processed independently from the raw data, pairwise similarity computation (cost of $O(D)$) is calculated across all windows, resulting in $O((N \times W) \times (N \times W) \times D)$ complexity. Subsequently, N data points are evenly divided into positive/negative samples, and all W^2 window pairs are compared for each positive–negative pair. Thus, the mining complexity is $O((N/2)^2 \times W^2)$.

In contrast, ESPRESSO performs online batch-hard mining within each batch while computing the loss. For each pair of samples in a B-sized batch, ESPRESSO compares similarity only between same-index window pairs rather than all W^2 window combinations. Therefore, the complexity of similarity calculation is $O(B^2 \times W \times D)$. Averaging similarities over all W yields a $B \times B$ similarity matrix, on which hard negative mining incurs a complexity of $O(B \times B)$. Although this process is repeated N/B times per epoch, the overall complexity of mining ($O(N/B \times (B \times B))$) and similarity calculation ($O(N/B \times (B^2 \times W \times D))$) remains significantly more efficient.

Summary. Overall, ESPRESSO remains lightweight and efficient due to its advanced architecture and mining strategies, requiring less than 0.8% and 0.3% of the per-epoch training time of DeepCorr and DeepCoFFEA, respectively (See Table 3).

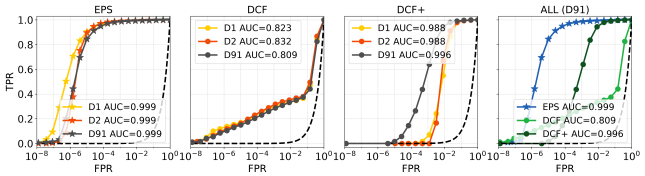
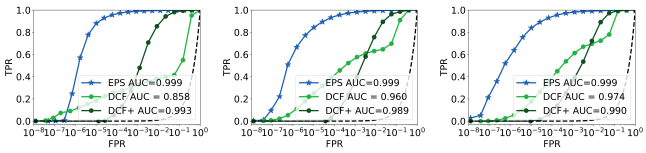


Figure 11: MLP classification performance of ESPRESSO and baselines under temporal drift. Trained with GTTCorr-Day1 and evaluated with GTTCorr-Day1, GTTCorr-Day2, and GTTCorr-Day91.



(a) Trained and evaluated with GTTCorr-All. (b) Trained with GTTCorr-All and evaluated with CellFlow-All-GT. (c) Trained with CellFlow-All-GT and evaluated with GTTCorr-All.

Figure 12: MLP classification performance of ESPRESSO and baselines in (a): the standard setting with identical training and test datasets, (b), (c): the network and long-term shift settings with different training and test datasets.

E MLP Classification Evaluation on Baselines

We explored MLP evaluation for ESPRESSO, DeepCoFFEA, and DeepCoFFEA+. As noted in Section 7.2, DeepCorr was excluded as its binary output is not suitable for MLP-based evaluation.

In the short-term temporal drift setting (Figure 11), similar trends to the thresholding evaluation (Section 7.3) are observed. The performance of all models remained stable, with only DeepCoFFEA showing a slight degradation of -0.014 AUC. Nevertheless, ESPRESSO consistently outperformed all baselines when FPRs are significantly low.

As illustrated in Figure 12, ESPRESSO achieved a high AUC of 0.999 across the standard and network/long-term shift settings. This indicates that ESPRESSO maintains strong performance regardless of the evaluation methodology. Similar to the thresholding evaluation results, DeepCoFFEA exhibited a sudden performance drop around FPR=10^{-1}, and DeepCoFFEA+ showed comparable performance to ESPRESSO. However, DeepCoFFEA+ still exhibited a sharp decrease in TPR within the range of $10^{-6} < \text{FPR} < 10^{-3}$, indicating limited real-world feasibility.

Across all settings, all models demonstrated relatively robust performance compared to DeepCorr, which can be attributed to the robustness of the metric learning approach. Also, it is notable that the ESPRESSO's ROC curve could cover the range of TPR=FPR=0 across various settings. Unlike discrete ranking-based thresholding evaluation, MLP classification produces continuous prediction probabilities, enabling more fine-grained performance assessment and providing a strong alternative or complementary approach to threshold-based evaluation.